

A Direct Junction Temperature Measurement Technique for Power LEDs

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Abstract—A study on power light-emitting diode (LED) junction temperature sensing based on the measurements of its forward voltage is presented. The linear dependence on temperature of the voltage drop across a power LED in the wide temperature range from $T = 35$ up to 175°C , for different bias currents, is investigated. The experimental measurements, repeated on three devices with the same parts number, exhibit a good degree of linearity. Moreover, the proposed sensors have excellent performances at the bias current of 1.4 mA with good repeatability and maintaining a stable output over more cycles of measurements.

Keywords— Temperature sensors, Temperature control, Light emitting diodes, Junction temperature.

I. INTRODUCTION

Light-emitting diodes (LEDs) are much more efficient than conventional light sources, such as incandescent and fluorescent light bulb. In fact, LEDs have a reduced power consumption with consequent increase of efficiency and lifetime. Despite their high efficiency, power LEDs dissipate a considerable amount of heat compared to their size. The resulting increase of junction temperature gives rise to various problems such as shorter lifetime, chromaticity changes and reduction of light output [1-3]. Consequently, these effects can be controlled by a constant monitoring and regulation of their temperature. Many methods have been proposed in the literature for monitoring the junction temperature based on LED electroluminescence [4], Raman spectroscopy [5,6], light spectrum [7], liquid crystal thermography [8] or on the use of external temperature sensors that estimate LED junction temperature.

All these methods require an additional external sensor for extraction of LED junction temperature. Another way for the thermal monitoring of power LED junction temperature consists in using electrical parameters e.g. reverse current [9]

and forward voltage under a constant current [10-13] that are temperature dependent.

In this work, a detailed study on the linear dependence on temperature of the voltage drop across a solid-state light source is presented. The sensor sensitivity, together with the linearity of output characteristics, are analyzed in a wide range of bias currents, and temperatures from 35 to 175°C . The experimental measurements, repeated on three devices with the same parts number, demonstrate the possibility to implement a method for monitoring and controlling the LED junction temperature in order to increase its lifetime and performances.

II. EXPERIMENTAL SETUP AND CHARACTERIZATION METHOD

As well known, the electrical characteristics of power LEDs are similar to that of diodes.

Moreover, a p-i-n, p-n or Schottky diode can operate as a temperature sensor when it is forward-biased by a low current I_F kept constant over the whole temperature working range [14-17]. The voltage drop appearing at the junction allows an indirect and accurate temperature measurement. Generally, the sensor is biased at low current values in order to avoid both self-heating effects, which can negatively affect the sensor linearity, and excessive power consumption.

As well known, the current I_F of a diode at a given applied voltage, V_F , can be modelled by the following formula:

$$I_F = I_S \left(e^{\frac{q(V - R_S I)}{\eta k T}} - 1 \right) \quad (1)$$

where η is the ideality factor, I_S is the saturation current, R_S is the series resistance, q is the electronic charge and k is the Boltzmann constant.

When the diode is forward biased and $q(V_F - R_S I_F) \gg \eta k T$, the voltage drop dependence on temperature can be obtained from (1):

$$V_F = \frac{kT}{q} \eta \ln \left(\frac{I_F}{I_S} \right) + R_S I_F \quad (2)$$

As demonstrated in [18], this relation shows a good linearity if the contribution of the series resistance can be considered negligible.

The experimental measurements are performed on commercial white power LEDs (XQAAWT-02-0000-00000B4E3) having a normal operation current of 175 mA with a forward voltage of 3 V. The maximum working temperature is of 175°C [19].

The starting point of the analysis was the identification of the best bias current range where LEDs are able to work as temperature sensors. In Fig.1 is shown a block diagram with the measurement system and the characterization method. The LEDs have been tested by increasing and decreasing the temperature in steps of 10 °C in a thermostatic oven (Fratelli Galli G-2100 [20]). The temperature measurements were repeated for three days on each single device to verify repeatability. The oven has a PT100 sensor, having an accuracy of 1 °C. Moreover, A PT100 temperature sensor (with an accuracy of 0.2 °C) has been placed on the printed circuit board (PCB) in contact with the device under test

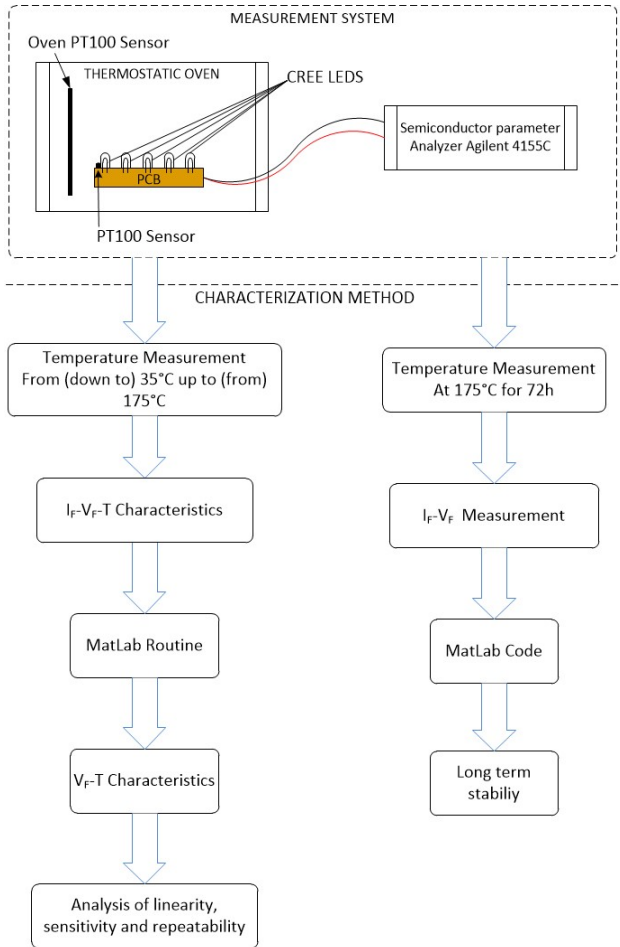


Fig.1. Measurement system and charatterization metod block diagram.

(DUT) in order to monitor, during measurements, the exact temperature set points, gradually varied from (down to) 35 °C up to (from) 175 °C.

The current-voltage (I_F - V_F) characteristics were obtained by connecting anode and cathode of each LED to an Agilent 4155C semiconductor parameters analyzer [21] and providing a bias current (I_F) from 10 μ A up to 10 mA. The corresponding voltage drop across the DUT was measured for the considered current and temperature ranges. At each temperature setting point, measurements started after waiting for the settling of stable temperatures for about three minutes.

The current-voltage-temperature I_F - V_F - T characteristics have been elaborated through a proper Matlab routine. In this way, from I_F - V_F - T measurements, the V_F - T characteristics have been extracted. Moreover, from the experimental data the linear fitting was calculated in order to evaluate the performances of the sensors in terms of linearity, sensitivity and root mean square error.

Subsequently, a thermal stress at 175°C was applied on the DUT for 72h, for verify the long-term stability of LEDs under test.

III. SINGLE LED EXPERIMENTAL RESULTS

A. Voltage-Temperature Characteristics

Measurements were performed on three LEDs with the same part number. Each DUT was characterized along three days. In our setup, the dc bias current, I_F , was varied in a range from 10 μ A to 10 mA and the corresponding voltage drop V_F across the LED was measured as reported in Fig. 2 for one sample.

In Fig.3 are shown voltage across the LED (V_F) versus the temperature (T) characteristics. As can be seen, the voltage drop decreases with increasing temperature in the temperature range from 35°C up to 175 °C. Moreover, in Fig.3, the experimental data are reported together with the linear fitting.

In our analysis, the coefficient of determination (R^2) [22] has been calculated to evaluate the agreement between the experimental measurements and their linear best fit, $f_L(T)$. In particular, R^2 allows to quantify the sensor linearity goodness by fitting the experimental data with a linear model. In addition, it can be calculated by the following formula:

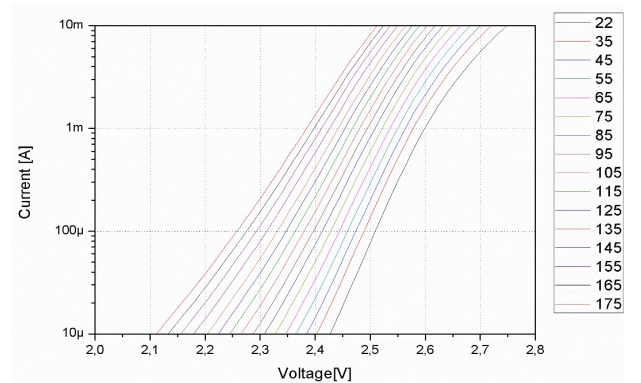


Fig.2. Currents-voltage characteristic of one analyzed sample.

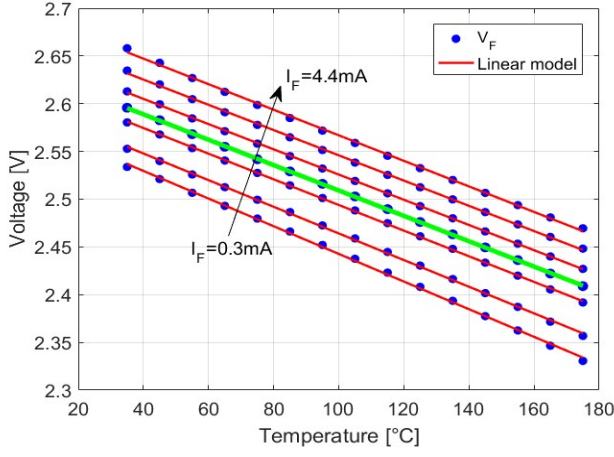


Fig.3. Measured (points) forward voltage versus temperature at seven currents I_F (0.3 mA, 0.5 mA, 1 mA, 1.4 mA, 2 mA, 3mA and 4.4 mA). Experimental data are fitted with the calculated linear model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (V_{F,i} - f_{L,i})^2}{\sum_{i=1}^n (V_{F,i} - \bar{V}_F)^2} \quad (3)$$

where n is the number of the experimental points, $V_{F,i}$ and $f_{L,i}$ are the i -values provided by the actual measure and by the best linear fit model, respectively, at the i -temperature, and $\bar{V}_F$ is the average of the all measurements.

Another important parameter for a temperature sensor linearity characterization is the temperature error, $rmse(T)$, between $f_L(T)$ and the experimental data, evaluated as:

$$rmse [K] = \frac{rmse [V]}{S} = \frac{\sqrt{\frac{\sum_{i=1}^n (V_{F,i} - f_{L,i})^2}{n}}}{S} \quad (4)$$

where $rmse(V)$ is the root mean square error of the diode voltage drop and S is the sensor sensitivity obtained from the slope of $f_L(T)$.

Table I reports a detailed description of coefficient of determination, sensitivity and $rmse$ for the V_F - T characteristics of Fig.1. All the sensor output characteristics, related to the diode taken into consideration, have a good linearity ($R^2 > 0.999$) for currents range from 0.3 mA up to 4.4 mA and for the wide temperature range between 35 °C

and 175 °C. The maximum linearity $R^2=0.99992$ is obtained for $I_F=1.4$ mA. Moreover, when I_F is 4.4 mA the sensitivity is 1.34 mV/°C and increases to 1.45 mV/°C at $I_F=0.3$ mA.

TABLE I. SENSOR PARAMETERS CALCULATED FROM EXPERIMENTAL DATA OF FIG. 1

Current [mA]	R^2	S [mV/°C]	rmse [mV]	rmse [°C]
0.3	0.99913	1.45	1.972	1.36
0.5	0.99947	1.39	1.473	1.06
1	0.99981	1.34	0.844	0.63
1.4	0.99992	1.33	0.519	0.39
2	0.99990	1.32	0.594	0.45
3	0.99959	1.32	1.227	0.93
4.4	0.99902	1.34	1.943	1.45

B. Evaluation of Sensor Best Bias Current

As previously described, the experimental measurements has been repeated, for each DUT, in three different days and for each day three measuring cycles were performed in a temperature range from 35 °C to 175 °C (and vice versa) with a current I_F from 10 μ A to 10 mA. Fig. 4 shows the linearity for the three characterized LEDs in the different measurement cycles. As can be seen, they have very

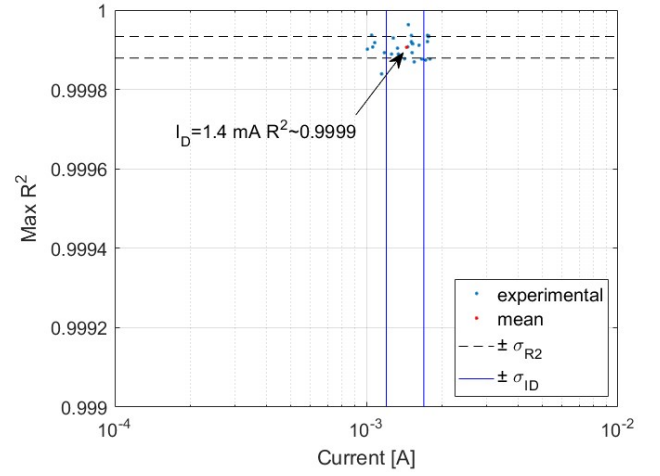


Fig.5. Maximum of R^2 versus current for different measurement days and devices.

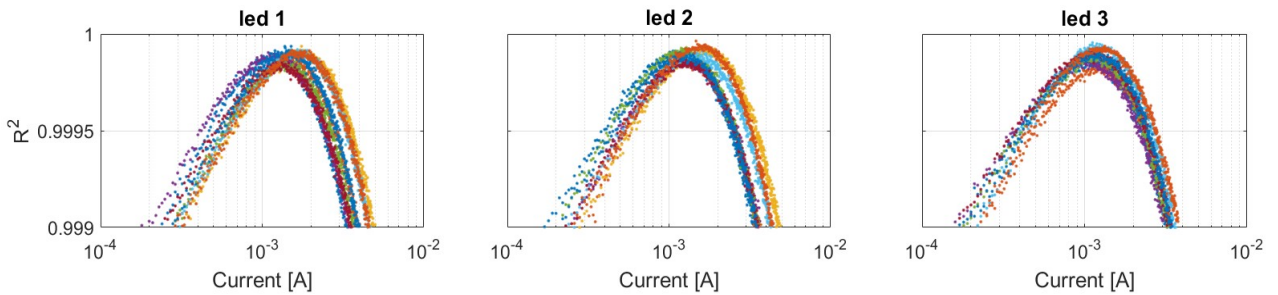


Fig.4. Coefficient of determination for three characterized LEDs.

repeatable performances.

For each curve, the maximum value of the determination coefficient has been extracted and in Fig. 5 are reported the calculated value. The dotted lines are the standard deviation of R^2 and the vertical continuous-lines (blue) are the standard deviation of the current I_F . Inside the square there are R^2 values for current range of $1.13 \text{ mA} < I_F < 1.67 \text{ mA}$. While the red dot is the average value of R^2 . In this point the mean value of the current is $I_{F_mean} = 1.4 \text{ mA}$, which corresponds to an average linearity value of about 0.9999. In conclusion, the value $I_{F_mean} = 1.4 \text{ mA}$ has been chosen as the best bias current that allows to use LEDs as temperature sensors.

C. Sensors performances at $I_F=1.4 \text{ mA}$

After evaluation of devices performances, the repeatability of three temperature sensors was verified at the best operating currents of $I_{F_mean}=1.4 \text{ mA}$ in different days and for different temperature ramps. The overall results are summarized in Fig. 6. In particular, at each temperature there are 27 points, corresponding to the measurements made in the above-mentioned three temperature ramps in three days, on three LEDs.

The experimental data, obtained from measurements, were modelled with the calculated linear best-fit $f_L(T)$ that can be described by the following equation:

$$V_F = -S \cdot T + Q = -1.36 \cdot 10^{-3} \cdot T + 2.64 \quad (5)$$

where Q is the intercept with the y-axis.

Another important parameter for the characterization of a temperature sensor is the temperature error. In this case, the absolute percentage error e_r of 27 experimental points was extracted from the data of Fig. 6 with respect to their linear best fit.

As shown in Fig. 7, e_r decreases with increasing temperature. For temperatures above 95°C , which might be considered as a safety threshold to avoid LED malfunctions, the error is less than 0.65 %. This appears to be a small

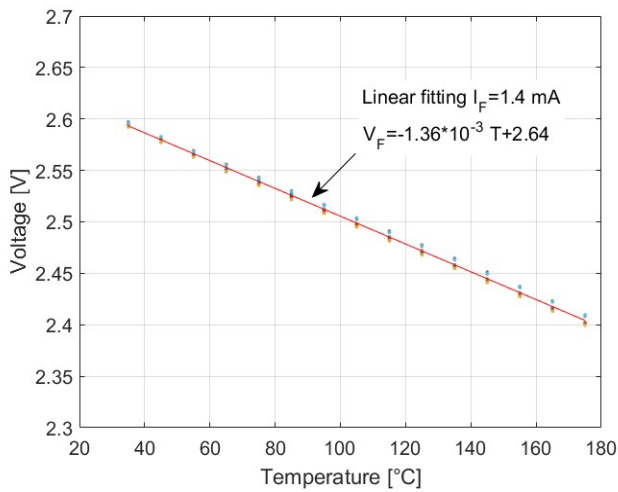


Fig.6. Linear fit $f_L(T)$ and experimental points of the V_F - T characteristics for three LEDs measured at the bias current of $I_{F_mean}=1.4 \text{ mA}$

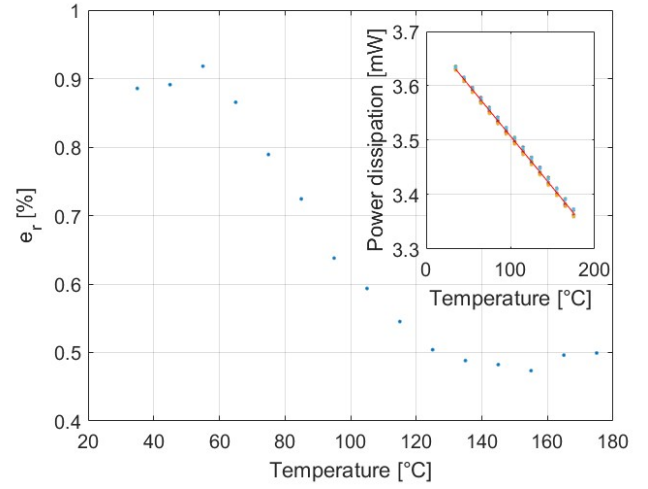


Fig.7. Absolute percentage error between the linear fitting and forward voltage (V_F) at bias current of $I_{D_mean}=24 \mu\text{A}$. The inset shows power dissipation versus temperature.

maximum error if we consider that we are directly measuring the LED temperature junction. The selected best current ($I_{F_mean} = 1.4 \text{ mA}$) is low with respect the normal operation LEDs current ($I_F=175 \text{ mA}$). This minimizes the self-heating effects on the extracted temperature. The inset of Fig. 7 shows the power dissipation of the temperature sensors in the considered temperature range. It decreases with temperature from a value of 3.6 mW at 35°C to 3.37 mW at 175°C . The corresponding temperature variation (ΔT) is equal to 0.072°C and 0.067°C , respectively. Such increase of the device temperature does not influence the sensor performances because it is about one order of magnitude lower than the minimum sensor temperature error.

D. Thermal Stability

In order to evaluate the thermal stability of the analyzed devices we have tested the drift of forward voltage V_F when

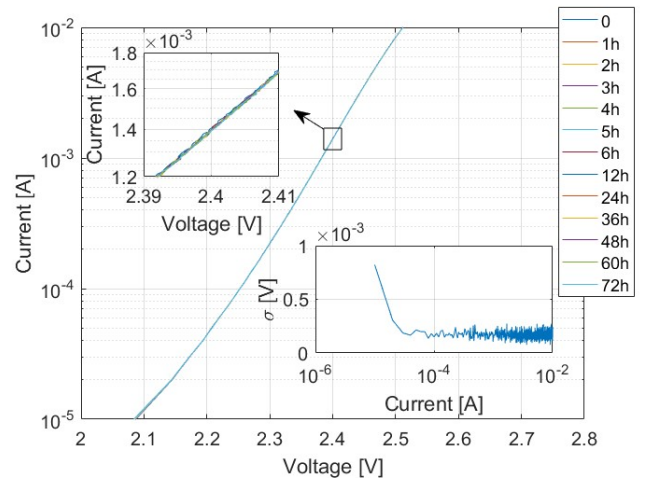


Fig.8. Current-voltage characteristics at $T=175^\circ\text{C}$. As an example, a detail of experimental experimental data is shown in the upper left inset. While in the lower right inset are reported the standard deviation σ versus current.

the LEDs are biased with a low constant current ($10 \mu\text{A} < I_F < 10 \text{ mA}$) with the DUTs continuously kept at 150°C for about 72 h.

The experimental results are summarized in Fig. 8. As shown in detail in the upper left inset (zoom of characteristics in the current range from 1.2 to 1.8 mA) the current-voltage characteristics are clearly overlapped.

Moreover, in order to evaluate the small variation present in the graph, the standard deviation was calculated. As reported in the lower right inset, the calculated standard deviation (σ) is very small, always lower than $8.2 \cdot 10^{-4} \text{ V}$.

IV. CONCLUSIONS

The work proposed a method for measuring the junction temperature of power LEDs used for lighting. It requires the forward biasing of the LED with a small current and measuring the voltage drop at its terminals. The experimental measurements have been performed on three commercial devices. The method shows both a high degree of linearity, with an average value of $R^2 = 0.9999$ at $I_{F_mean} = 1.4 \text{ mA}$, and a good sensitivity ($S = 1.36 \text{ mV}/^\circ\text{C}$) in the temperature range $T = 35\text{--}175^\circ\text{C}$. Different cycles of measurements have shown a good long-term stability and output repeatability.

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